Assignment: Git & GitHub Fundamentals

Objective-:

The purpose of this assignment is to evaluate your understandin
Control, and basic Git workflows used in software development.

Part A: Theory Questions

1. What problem does Version Control solve in software develop

Version Control helps developers track changes in files. It preve

2. Define Version Control System (VCS).

A Version Control System (VCS) is a software tool that records c

3. What is Git?

Git is a free and open-source Distributed Version Control Systen

4. Explain the difference between Git and GitHub.

Git is a Version Control System. GitHub is an online platform for I
Works locally on your computer. Works on the internet (cloud).

Tracks file changes. Stores and shares Git repositories.

5. What is a Repository?

A Repository (Repo) is a storage location where all project files, f

6. What is a Commit in Git?

A Commit is a snapshot of the project at a particular point in time

7. What is the purpose of the Staging Area?

The Staging Area is used to select the files that will be included i

8. Explain the following Git architecture components.
Working Directory

The place where you create and edit project files.

Staging Area

A temporary area where changes are prepared before committir

Local Repository

The Git repository stored on your own computer.

Remote Repository

The online copy of the repository stored on platforms like GitHul

9. What is the purpose of the following commands?

a) git init

Initializes a new Git repository.

b) git status

Shows the current status of files.

c) git add .

Stages all changed files.

d) git commit -m "message"

Creates a commit with a message.

e) git push

Uploads local commits to GitHub.

10. Explain the industry workflow of Git from cloning a project to

Clone the repository.
Create or edit files.
Check status.
Stage changes.
Commit changes.
Pull latest updates.
Push changes to GitHub.

11. What information is stored inside a Git commit?

A Git commit stores:

Changed files
Commit message
Author name
Author email
Date and time
Unique Commit ID (Hash)

12. Why is Git considered a Distributed Version Control System?

Because every developer has a complete copy of the repository a

13. What are the advantages of using GitHub?

Online backup
Easy collaboration
Project sharing
Issue tracking
Pull Requests
Free hosting for Git repositories
14. What is the difference between Push and Pull?

Uploads local changes to GitHub. Downloads latest
changes from GitHub.

15. Explain the role of Git in team collaboration.

Git allows multiple developers to work on the same project, mana

Part B : Practical Assignment
Task 1 : Git Configuration

```
To https://github.com/AnvitaChauhan977/Learning-With-CodroidHub.git
   b09bf28..e44c4b4  daily-practice -> daily-practice
PS C:\Users\hp\OneDrive\Desktop\Learning-With-CodroidHub>
 *  History restored
                                                git --versions\hp\OneDrive\Desktop\Learning-With-CodroidHub>
git version 2.54.0.windows.1
PS C:\Users\hp\OneDrive\Desktop\Learning-With-CodroidHub> git config --global user.name "Anvita Chauhan"
PS C:\Users\hp\OneDrive\Desktop\Learning-With-CodroidHub> git config --global user.email "your-anvita.chauhan_cs24@gla.ac.in"
PS C:\Users\hp\OneDrive\Desktop\Learning-With-CodroidHub> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=Anvita Chauhan
user.email=your-anvita.chauhan_cs24@gla.ac.in
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/AnvitaChauhan977/Learning-With-CodroidHub.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
branch.daily-practice.vscode-merge-base=origin/main
branch.daily-practice.remote=origin
branch.daily-practice.merge=refs/heads/daily-practice
PS C:\Users\hp\OneDrive\Desktop\Learning-With-CodroidHub>
```

Task 2 : Create Local Repository

TASK 2: CREATE LOCAL REPOSITORY

1. mkdir git-assignment

```
Anvita@DESKTOP-7F5J385 MINGW64 ~
$ mkdir git-assignment
```

2. cd git-assignment

```
Anvita@DESKTOP-7F5J385 MINGW64 ~
$ cd git-assignment

Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment
```

3. echo. > index.html

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment
$ echo. > index.html
```

4. git init

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment
$ git init
Initialized empty Git repository in C:/Users/Anvita/git-assignment/.git/
```

5. git status

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git status
On branch master

No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html
nothing added to commit but untracked files present (use "git add" to track)
```

TASK 3: FIRST COMMIT

1. git status

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

nothing added to commit but untracked files present (use "git add" to track)
```

2. git commit -m "Initial Commit"

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git add .
$ git commit -m "Initial Commit"
[master (root-commit) a5e3b6f] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 index.html
```

3. git log

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git log
commit a5e3b6f3e7e8c3d4e2f7d9c6b1a2f3d4e5f6a7b8
Author: Anvita Chauhan <anvita123@gmail.com>
Date:   Sun Jun 23 21:45:10 2024 +0530

    Initial Commit
```

TASK 4: MODIFY AND COMMIT

1. git status

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

2. git add .

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git add .
```

3. git commit -m "Added Student Name"

```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git commit -m "Added Student Name"
[master 2b7d9e1] Added Student Name
 1 file changed, 1 insertion(+)
```

4. git log

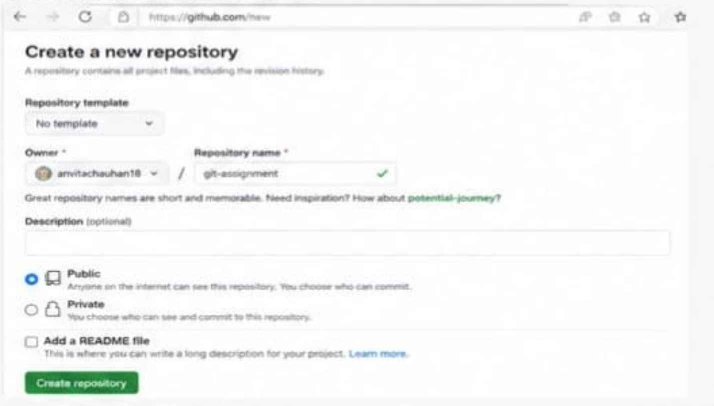
```
Anvita@DESKTOP-7F5J385 MINGW64 ~/git-assignment (master)
$ git log
commit 2b7d9e1c4a9f9b0d7e6c5b4a3d2e1f0a9b8c7d6e
Author: Anvita Chauhan <anvita123@gmail.com>
Date:   Sun Jun 23 21:50:22 2024 +0530

    Added Student Name

commit a5e3b6f3e7e8c3d4e2f7d9c6b1a2f3d4e5f6a7b8
Author: Anvita Chauhan <anvita123@gmail.com>
Date:   Sun Jun 23 21:45:10 2024 +0530

    Initial Commit
```

TASK 5: CREATE GITHUB REPOSITORY



TASK 7: PUSH PROJECT TO GITHUB

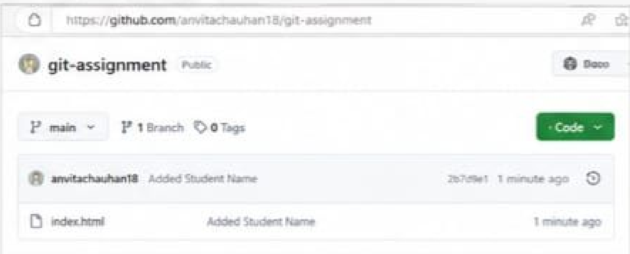
1. git branch -M main

```
Anvita@DESKTOP-7F5J985 MINGW64 ~/git-assignment (master)
$ git branch -M main
```

2. git push -u origin main

```
Anvita@DESKTOP-7F5J98S MINGW64 ~/git-assignment (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 308 bytes | 308.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/anvitachauhan18/git-assignment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

3. GitHub repository after push

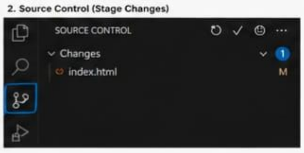


TASK 8: VS CODE GIT OPERATIONS

1. Modify index.html

```
index.html
1  <h1>My First Git Project</h1>
2
3  <p>Anvita Chauhan</p>
4
5  <p>Learning Git and GitHub</p>
```

2. Source Control (Stage Changes)



3. Commit Changes

4. Push Changes